



Pharm*Achieve*

DRUG APPROVAL PROCESS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

Knowledge
& Expertise

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

ADR	Adverse Drug Reaction
ANDS	Abbreviated New Drug Submission
BID	Bis In Die (Twice Daily)
CADTH	Canadian Agency for Drugs and Technologies in Health
CDA	Canada's Drug Agency
CDEC	Canadian Drug Expert Committee
CDR	Common Drug Review
DIN	Drug Identification Number
DIN-HM	Drug Identification Number-Homeopathic Medicine
DPD	Drug Product Database
F&DA	Food and Drugs Act
F/P/T	Federal, Provincial, or Territorial
HPFB	Health Products and Food Branch
IBS	Irritable Bowel Syndrome
MAC	Medical Advisory Committee

MR	Modified-Release
NDS	New Drug Submission
NHP	Natural Health Product
NOC	Notice of Compliance
NOC/c	Notice of Compliance with Conditions
NPN	Natural Product Number
OBGYN	Obstetrician-Gynecologist
P&T	Pharmacy and Therapeutics
pCODR	Pan-Canadian Oncology Drug Review
PERC	pCODR-Expert Review Committee
PDD	Pharmaceutical Drugs Directorate
PLA	Product License Application
PMPRB	Patented Medicine Prices Review Board
PO	Per Os (Oral)
SAP	Special Access Program

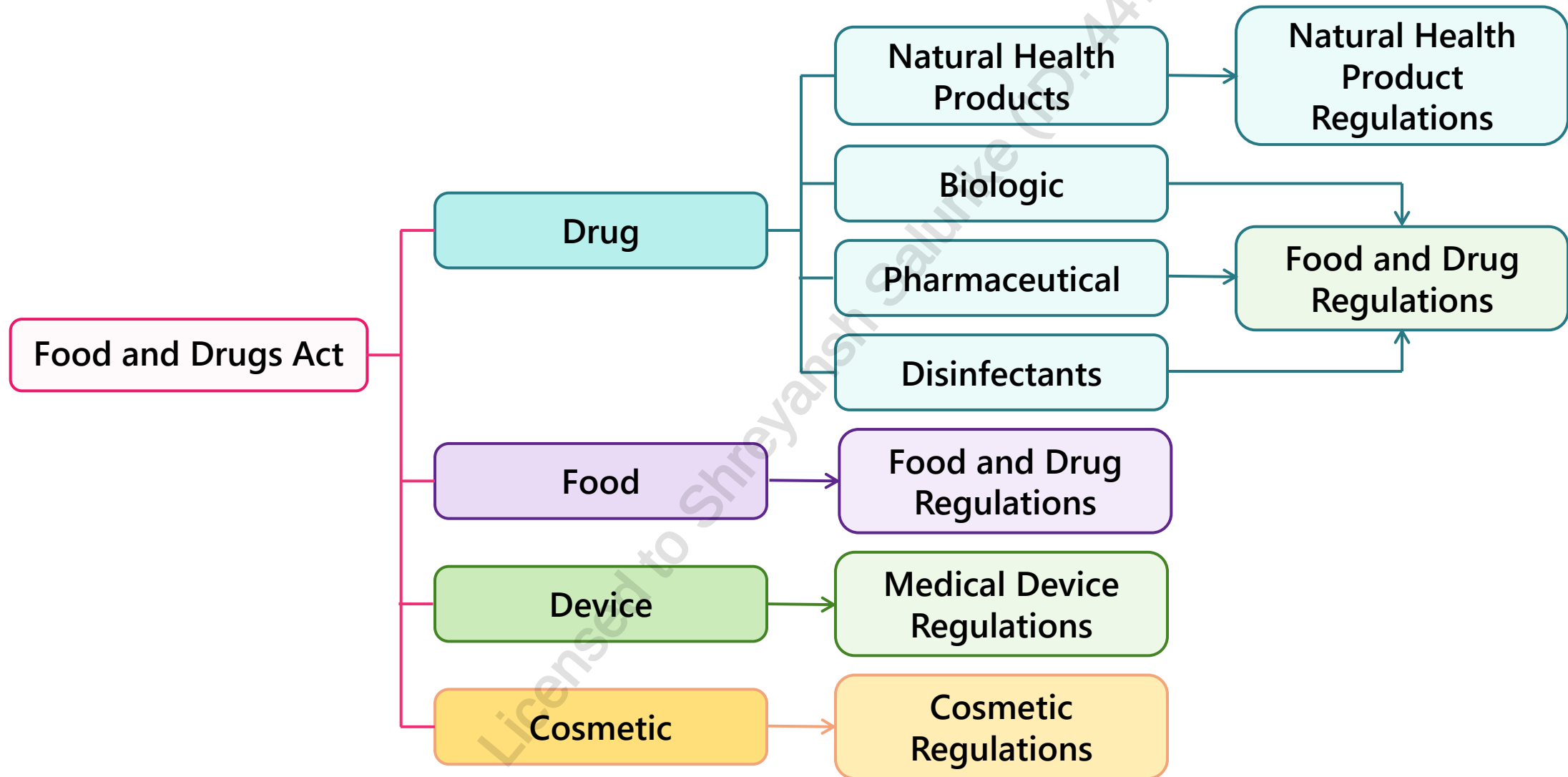
OBJECTIVES

- 1 Describe the role of Health Canada in the drug approval process
- 2 Describe the role of the Common Drug Review (CDR) and the Canadian Drug Agency (CDA), formerly Canadian Agency for Drugs and Technology in Healthcare (CADTH)
- 3 Explain the role of the provincial ministry in drug approval for the provincial formulary
- 4 Understand the role of the Pharmacy and Therapeutics (P&T) Committee in the drug approval process for hospital formularies

ROLE OF HEALTH CANADA

- Health Canada is the **federal body** responsible for...
 - Ensuring pharmaceutical products available to Canadians are **safe, effective, and of high-quality**
 - Ensuring that pharmaceutical products **comply with Food and Drugs Act & Regulations**
- It has various departments overlooking specific activities of the drug approval process
- An application of approval is submitted to the Health Products and Food Branch (HPFB) of Health Canada and then to its respective department depending upon the classification of product (e.g. Pharmaceutical Drug Directorate, Medical Devices Directorate, Natural and Non-prescription Health Products Directorate, etc.)
- Classification of the product according to Food and Drugs Act (F&DA) is the first regulatory step of the approval process

CLASSIFICATION OF PRODUCTS UNDER FOOD & DRUGS ACT



DRUG APPROVAL PROCESS

STEP 1- APPLICATION SUBMISSION

Manufacturer makes an application for a **New Drug Submission (NDS)**

- Apply for specific indication
- Pay application fee
- If it is a breakthrough drug for life-threatening or debilitating conditions, then priority review may be given

Manufacturer submits evidence from clinical trials

- **Safety** data → Is the drug safe? Serious harm? Side effects?
- **Efficacy** data → Does the drug work in improving outcomes, symptoms, or curing disease?
- Do the **benefits outweigh the risks**? → For life-threatening diseases (e.g. cancer therapy) or for conditions with limited or no other treatment options

DRUG APPROVAL PROCESS

STEP 2- DRUG REVIEW

Applications of pharmaceutical drugs for human use are reviewed by Pharmaceutical Drugs Directorate (PDD) for its efficacy, safety, quality, and regulatory standards.

DRUGS

Include:

- Prescription products
- Biologically derived vaccines*
- Blood products, serums, tissues, and organs*
- Radiopharmaceuticals*

**Biologic and Radiopharmaceuticals Drugs Directorate regulates biological drugs, radiopharmaceuticals, and cells/tissues/ organs for human use*

Defined as:

Substances used in diagnosis, treatment, mitigation, or prevention of disease in humans or animals

ROLE OF THE PHARMACEUTICAL DRUGS DIRECTORATE (PDD)

- Federal authority under Health Products and Food Branch (HPFB) of Health Canada
- Regulates, evaluates and monitors safety, efficacy and quality of therapeutic and diagnostic products available in Canada
- Reviews NDS (New Drug Submission)
- Provides Notice of Compliance (NOC) once all issues have been resolved
- Provides Drug Identification Number (DIN)
- Regulates product labelling (to avoid errors)
- Monitors ADRs and toxicities (for safety)
- Review Special Access Program (SAP) requests

DRUG APPROVAL PROCESS

STEP 3- NOC AND DIN PROVIDED

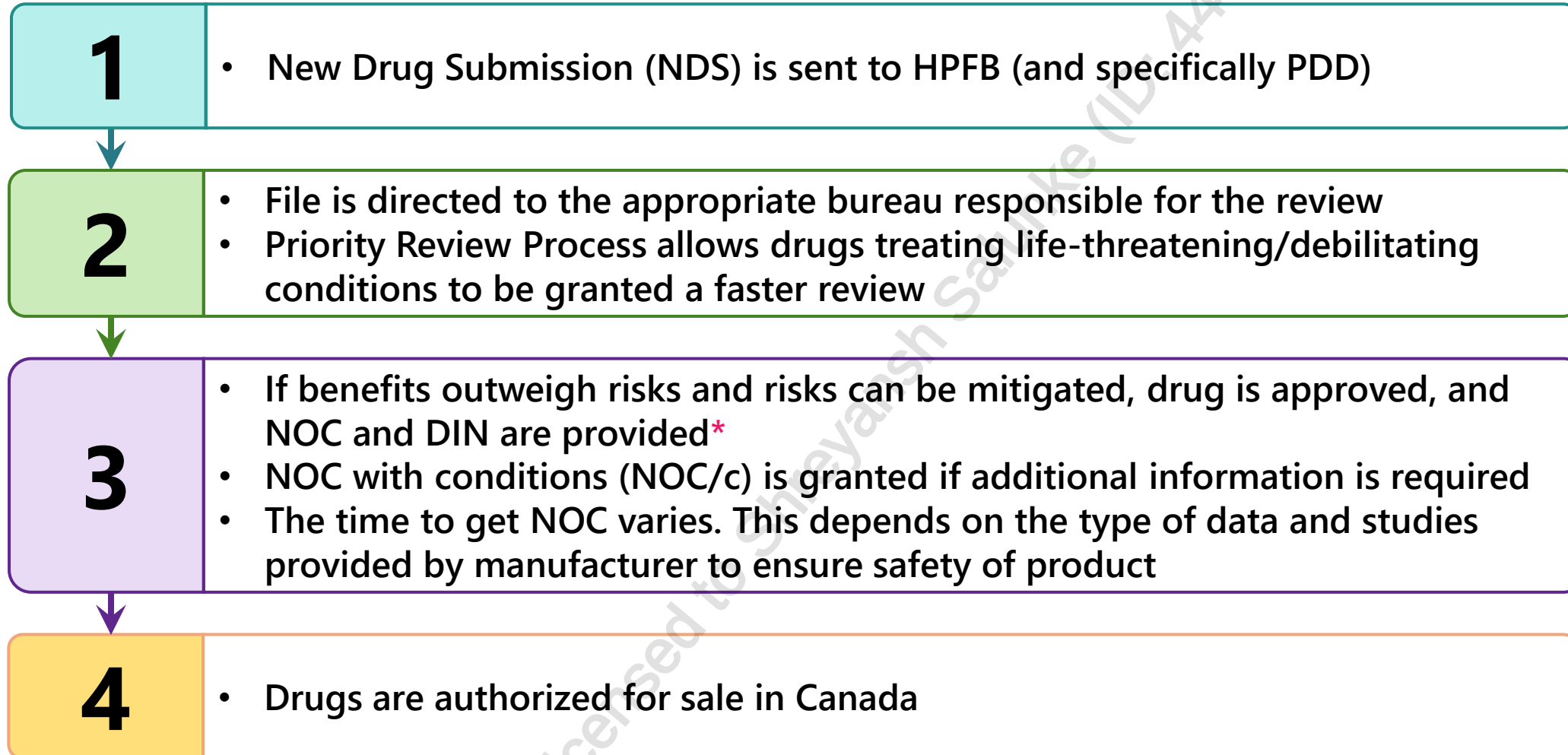
Notice of Compliance (NOC)

- A document from Health Canada indicating that the drug meets Food and Drugs Act requirements and gives permission to market the drug in Canada

Drug Identification Number (DIN)

- A unique number printed on the label that identifies the specific combination of active ingredient, brand name, strength (dose), route of administration, manufacturer, and dosage form

DRUG APPROVAL PROCESS SUMMARY



**Following or before approval, certain biological products may be tested through Lot Release Process to monitor for safety, efficacy, and quality*

NATURAL AND NON-PRESCRIPTION HEALTH PRODUCTS DIRECTORATE (NNHPD)

Federal Authority under HPFB that monitors sale of natural health products and non-prescription drugs which includes disinfectant drugs.

Natural health Product (NHP)

Substance that occurs naturally and is used to maintain or restore health

- Vitamins and minerals
- Herbal remedies
- Traditional medicines (Chinese/Ayurvedic)
- Homeopathic medicines
- Probiotics
- Other products (amino acids, essential fatty acids, etc.)

Non-prescription Drug

Pharmaceutical drug products that can be purchased without a prescription

- Over-the-counter drugs: analgesics, cough and cold remedies, laxatives, antacids
- Behind-the-counter drugs: emergency birth control drugs, certain heartburn drugs
- Disinfectants: hard surface disinfectants (e.g. for countertops)

NATURAL HEALTH PRODUCTS (NHPs)

- Subject to the [Natural Health Products Regulations](#)
- All NHPs must have product licenses. This can be obtained through a Product License Application (PLA) for Natural Health Products.
- Health Canada assesses the product for safety, efficacy, and quality. If it meets the criteria, it is given a product license and Natural Products Number (NPN) or Drug Identification Number-Homeopathic Medicine (DIN-HM).
- Canadian sites that manufacture, package, label, and import NHPs must have site licenses.
- Healthcare practitioners who compound NHPs for their patients or retailers of NHPs do not require a site license.
- Product license holders must monitor adverse reactions and report all serious ones to Health Canada

LABELLING OF NHPs

All NHPs must have labels which include

- Product name
- Product license number
- Quantity of product in bottle
- Lot number
- Complete list of medicinal and non-medicinal ingredients
- Recommended use (purpose/health claim, route and dose)
- Any cautionary statements, warnings, contraindications and possible adverse reactions
- Any special storage conditions

Principle display panel labelling requirements

- Primary brand name
- Product number (NPN)
- Dosage form
- The words 'sterile' and 'stérile' if it is sterile
- Net amount in immediate container (weight, measure or number)

FYI

Natural Health Products Regulations were amended in 2022 and brought about changes to NHP labelling requirements including the addition of priority allergen labelling, standardized product facts table, more modern contact information (e.g. website), improved contrast, and increased font. Existing products have until June 22, 2028, to amend their packaging. Products approved June 21, 2025, or after need to abide by these requirements immediately.

CANADA'S DRUG AGENCY (CDA, *formerly* CADTH)

- Independent, unbiased organization that provides evidence on drugs, diagnostic tests, devices and procedures to healthcare decision-makers
- It seeks advice from [Canadian Drug Expert Committee \(CDEC\)](#) for non-cancer drugs, and the [pan-Canadian Oncology Drug Review \(pCODR\) Expert Review Committee \(pERC\)](#) for cancer drugs
- These committees consist of an appointed, national, independent body of physicians, pharmacists, other health care professionals, and public members who evaluate clinical and pharmacoeconomic data. They make recommendations to participating federal, provincial, and territorial drug plans on whether drugs should be listed for coverage. They also advise on decision-making and strategies for optimal drug use.

COMMON DRUG REVIEW (CDR)

- National review in which the drug sponsor requests a **Common Drug Review (CDR)** or **pan-Canadian Oncology Drug Review (pCODR)** to have the patented drug covered by publicly funded drug plans
- CDR submission eligibility:
Any new drug, combination products, or drugs with new indications that have received NOC, NOC with conditions (NOC/c) or new drugs with a pending NOC or NOC/c. Natural Health Products are ineligible
- Administered by **CDA**

1. Review evidence comparing drug to currently available alternatives (where available) in terms of efficacy/safety. Also review patient group submissions to assess unmet needs



2. Analyze cost-effectiveness: is this drug cost-effective relative to available alternatives and in the context of the Canadian public health care system?



3. Recommendations are released to the public on the CDA website

- Final reimbursement and formulary decisions are made by participating drug plans (F/P/T)

PROVINCIAL MINISTRY DRUG REVIEW

A recommendation by CDR to list the drug on the formulary does not mean that the drug gets listed on the provincial formulary, as each province makes its own decision based on their therapeutic advisory committees. All requests for approval must be forwarded to the Cabinet Minister for final approval

Provincial/territorial ministries conduct their *own* reviews; building on top of work done by Health Canada and the CDR process

- Factors considered: mandate, health care funding priorities, substantial economical benefits, meeting a significant need, showing a significant therapeutic benefit, resources available, needs of patient advocacy groups
- Each province and territory is responsible for communicating with the manufacturer regarding the coverage status of the drug and listing decision

Generic drugs **do not** have to go through the therapeutic advisory committees for review to expedite process

- However, a general submission is still required because it needs be forwarded to the government for approval. This is called an Abbreviated New Drug Submission (ANDS)

KEY STEPS/STAKEHOLDERS TO DRUG ACCESS IN CANADA

Health Canada

- Approval of drug to market via NOC

CDA Review (CDR, pCODR)

- Cost-effectiveness and safety evaluation prior to drug being listed on any publicly funded formulary
- Provides recommendations for listing (or for not listing)

Provincial Drug Plans

- Decision based on the impact on the drug budget, need for access to drug therapy (province-dependent), comparative costs, patient advocacy groups, substantial economic or therapeutic benefits

KEY STEPS/STAKEHOLDERS TO DRUG ACCESS IN CANADA

Private Payer Coverage (Insurance Drug Plans)

- Insurance companies have separate review processes for medications (usually high-cost medications), which are required to ensure the long-term sustainability of drug plans
- Pharmaceutical companies also submit medication dossiers to insurance companies for review

Patented Medicine Prices Review Board (PMPRB)

- Protects Canadians by ensuring prices of patented medicines in Canada are not excessive

Hospitals

- Conduct own drug reviews and make hospital drug formulary decisions

PHARMACY & THERAPEUTICS (P&T) COMMITTEE

MISSION

Provide safe, rational, and cost-effective prescribing, distribution and administration of drugs and therapeutic agents **within the hospital setting**

TERMS OF REFERENCE: document issued by hospital outlining the purpose, limits and authority of a committee (usually issued by a higher authoritative body)

Mandates include:

- Review and revise guidelines/protocols such as administration of medications
- Review use of approved medications with off-label indications
- Support educational programs regarding hospital protocols/procedures to reduce risks of medication-related harms
- Review adverse drug reaction reports and make appropriate recommendations
- Perform other duties as directed by the corporate Medical Advisory Committee (MAC)
- Serve as an advisory body to members of the hospital on all matters pertaining to the use of drugs
 - Review and make recommendations to members of the hospital on policies and procedures relative to the safe, effective, and economical use of medications
- Maintain a current drug formulary and communicate formulary changes to hospital members

MEMBERS OF P&T COMMITTEE

Members of the P&T Committee come from a variety of disciplines (e.g. pharmacists, nurses, physicians, dietitians)

- Emergency services
- Anesthesiology
- Psychiatry
- Pharmacy
- Director of Pharmacy
- Clinical Specialist
- Nursing
- Nursing leadership
- Medicine
- Surgery or critical Care
- Emergency medicine
- Ad Hoc members
 - Dietitian
 - Chair – physician
- Co-Chair – pharmacy
- Administration
- VP from patient care area
- Director from patient care area
- Family Practice
- Internal Medicine
- Cardiologist, intensivist, medicine
- Infectious Disease
- Pediatrician
- OBGYN
- Surgery



Meeting Frequency: regularly
e.g. monthly meetings
(median: 9 times/year)

Agenda: distributed to
members a week in advance

Minutes: approved by Medical
Advisory Committee (MAC)
and is available to members
after approval

REQUEST FOR FORMULARY ADDITION

✓ THERAPEUTIC ASSESSMENT

- High-quality, large, RCTs showing therapeutic advantage over available drugs
 - Questionable/marginal improvement in patient outcomes
 - Efficacy advantage is somewhat offset by a toxicity disadvantage
 - Limitation in trial design does not allow for interpretation of data
- No therapeutic advantage but secondary characteristics of the drug confer some advantage over existing product
 - E.g. Dosage forms, frequency of administration, pharmacokinetics, drug interactions, safety profile, cost-effectiveness

✓ DRUG UTILIZATION

- Population therapeutics of the drug
- Expected place in therapy (1st line therapy, augmentation, refractory) and restrictions

✓ PROJECT ANNUAL COST IMPACT

- # of patients per year
- # of doses per year
- Acquisition cost per dose and indirect costs (e.g. nursing time, preparation)

✓ IMPLEMENTATION ISSUES

- Newsletter, pre-printed orders, parenteral monograph, education plan

✓ APPROVAL AND SIGNATURES

- Chief of department, nursing council, and other departments (if applicable)

✓ PHARMACOKINETIC AND/OR PHARMACODYNAMIC PARAMETERS

✓ HEALTH CANADA APPROVED INDICATIONS

✓ RECOMMENDED DOSING AND DURATION OF THERAPY

✓ COMPARATORS

✓ FORMULARY DRUG OR TREATMENT THAT THE REQUESTED DRUG WOULD REPLACE

✓ ODB AND MARKET AVAILABILITY

✓ SUMMARY OF CLINICAL EVIDENCE

✓ ASSESSMENT OF SIDE EFFECTS, SAFETY ISSUES, OR CONTRAINDICATIONS OF CONCERN

✓ PHARMACOECONOMIC ANALYSIS

✓ FOLLOW-UP EVALUATION

TAKEAWAY

- Drug Approval Process:
 1. Application submission: Manufacturer makes an application for a New Drug Submission (NDS) to HPFB
 2. Drug review: by PDD
 3. NOC and DIN provided
 4. Authorized for sale
- NHPs receive a NPN or a DIN-HM
- **CDA**: provides evidence on drugs, diagnostic tests and devices and procedures to healthcare decision-makers
- **CDEC**: advisory board to CDA
- **P&T Committee**: provides safe, rational, and cost-effective prescribing, distribution and administration of drugs and therapeutic agents within the hospital setting

CASE STUDY

UF is a 34-year-old male who presents to the community pharmacy with a prescription from the UK for mebeverine modified-release (MR) 200 mg PO BID to manage his irritable bowel syndrome (IBS) symptoms. UF recently moved to Canada 3 months ago and has been using a supply from the UK, however, he has now run out. He explains that he has tried other anti-spasmodic agents licensed in Canada, however, these medications were discontinued due to poor tolerability. After searching Health Canada's Drug Product Database (DPD), you notice that mebeverine is not available in Canada as there is no record of it. UF asks you if you can order mebeverine for him to manage his symptoms of abdominal cramping, pain, and spasmodic bowel contractions associated with IBS. He explains that he has been taking this medication for the past 5 years and it has provided a significant benefit in controlling his symptoms. He is keen on maintaining the same medication, however, he is open to trying an alternative based on your recommendation. You decide to discuss his case with UF's registered family physician.

CASE STUDY - QUESTIONS

1. UF asks you why this medication is not available in Canada. Which of the following explanations is the most accurate to provide to UF?
 - a) Mebevrine has been banned in Canada due to safety concerns
 - b) Mebevrine has not received a Notice of Compliance (NOC) from Health Canada
 - c) Mebevrine's patent has expired and it cannot be marketed in Canada
 - d) Mebevrine is available over-the-counter (OTC) only in Canada
2. Which of the following actions are Canadian pharmacists authorized to do?
 - a) Import mebevrine from the UK for UF
 - b) Substitute mebevrine with a similar drug marketed in Canada based on product monographs
 - c) Recommend a therapeutic alternative to mebevrine and inform the prescriber
 - d) Request access through Health Canada's Special Access Program (SAP)
3. Which of the following methods is the most appropriate for a Canadian physician to follow to obtain access to mebevrine for UF?
 - a) Request access through the Special Access Program (SAP)
 - b) Submit a New Drug Submission (NDS)
 - c) File an Abbreviated New Drug Submission (ANDS)
 - d) Apply for provincial formulary coverage
4. Which of the following statements is true regarding Health Canada's role in medication access?
 - a) Health Canada oversees and enforces the Common Drug Review (CDR)
 - b) Health Canada regulates product labeling
 - c) Health Canada assigns a Drug Identification Number (DIN) for natural and alternative medications
 - d) Health Canada determines public drug coverage under provincial insurance plans

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CHANGE LOG

July/August 2025

- Formatting and grammar changes throughout
- Slide 3: Updated legend
- Slides 24, 25: Case study and questions added

October 2025

- No content changes made

March 2026

- Slide 2: New NAPRA competencies added